

Exercise 8

Solution 1: In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$ [Sum of the angles of a triangle]

$$\Rightarrow \angle A + 76^\circ + 48^\circ = 180^\circ$$

$$\Rightarrow \angle A + 124^\circ = 180^\circ$$

$$\Rightarrow \angle A = 56^\circ$$

Solution 2: Let the angles of the given triangle measure $(2x)^\circ$, $(3x)^\circ$ and $(4x)^\circ$, respectively.

Then,

$$2x + 3x + 4x = 180^\circ \text{ [Sum of the angles of a triangle]}$$

$$\Rightarrow 9x = 180^\circ$$

$$\Rightarrow x = 20^\circ$$

Hence, the measures of the angles are $2 \times 20^\circ = 40^\circ$, $3 \times 20^\circ = 60^\circ$ and $4 \times 20^\circ = 80^\circ$.

Solution 3: Let $3\angle A = 4\angle B = 6\angle C = x$.

Then,

$$3\angle A = x$$

$$\Rightarrow \angle A = \frac{x}{3}$$

$$4\angle B = x$$

$$\Rightarrow \angle B = \frac{x}{4}$$

$$6\angle C = x$$

$$\Rightarrow \angle C = \frac{x}{6}$$

$$\text{Now, } \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \frac{x}{3} + \frac{x}{4} + \frac{x}{6} = 180^\circ$$

$$\Rightarrow \frac{4x + 3x + 2x}{12} = 180^\circ$$

$$\Rightarrow \frac{9x}{12} = 180^\circ$$

$$\Rightarrow 9x = 180^\circ \times 12$$

$$\Rightarrow x = 180^\circ \times \frac{12}{9}$$

$$\Rightarrow x = 20^\circ \times 12$$

$$\Rightarrow x = 240^\circ$$

$$\therefore \angle A = \frac{240^\circ}{3} = 80^\circ, \angle B = \frac{240^\circ}{4} = 60^\circ \text{ and } \angle C = \frac{240^\circ}{6} = 40^\circ$$

Solution 4: $A + B = 108^\circ$ [Given]

But as $\angle A$, $\angle B$ and $\angle C$ are the angles of a triangle,

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 108^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 108^\circ$$

$$\Rightarrow \angle C = 72^\circ$$

Also, $\angle B + \angle C = 130^\circ$ [Given]

$$\angle B + 72^\circ = 130^\circ$$

$$\Rightarrow \angle B = 130^\circ - 72^\circ$$

$$\Rightarrow \angle B = 58^\circ$$

Now as, $\angle A + \angle B = 108^\circ$

$$\angle A + 58^\circ = 108^\circ$$

$$\Rightarrow \angle A = 108^\circ - 58^\circ$$

$$\Rightarrow \angle A = 50^\circ$$

Hence, $\angle A = 50^\circ$, $\angle B = 58^\circ$ and $\angle C = 72^\circ$.

Solution 5: Let $\angle A + \angle B = 125^\circ$ and $\angle A + \angle C = 113^\circ$

Then,

$$\angle A + \angle B + \angle A + \angle C = (125 + 113)^\circ$$

$$\Rightarrow (\angle A + \angle B + \angle C) + \angle A = 238^\circ$$

$$\Rightarrow 180^\circ + \angle A = 238^\circ \quad [\because \angle A + \angle B + \angle C = 180^\circ \text{ (sum of angles of a triangle)}]$$

$$\Rightarrow \angle A = 58^\circ$$

$$\begin{aligned}\therefore \angle B &= 125^\circ - \angle A \\ &= (125 - 58)^\circ \\ &= 67^\circ\end{aligned}$$

$$\begin{aligned}\therefore \angle C &= 113^\circ - \angle A \\ &= (113 - 58)^\circ \\ &= 55^\circ\end{aligned}$$

Solution 6:

$$\begin{aligned}\angle P - \angle Q &= 42^\circ \text{ and } \angle Q - \angle R = 21^\circ \\ \Rightarrow \angle P &= 42^\circ + \angle Q \text{ and } \angle R = \angle Q - 21^\circ\end{aligned}$$

Now,

$$\angle P + \angle Q + \angle R = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\therefore 42^\circ + \angle Q + \angle Q + \angle Q - 21^\circ = 180^\circ$$

$$\Rightarrow 3\angle Q = 159^\circ$$

$$\Rightarrow \angle Q = 53^\circ$$

$$\therefore \angle P = 42^\circ + \angle Q = (42 + 53)^\circ = 95^\circ$$

$$\therefore \angle R = \angle Q - 21^\circ = (53 - 21)^\circ = 32^\circ$$

Solution 7: Let $\angle A + \angle B = 116^\circ$ and $\angle A - \angle B = 24^\circ$

Then,

$$\therefore \angle A + \angle B + \angle A - \angle B = (116 + 24)^\circ$$

$$\Rightarrow 2\angle A = 140^\circ$$

$$\Rightarrow \angle A = 70^\circ$$

$$\begin{aligned}\therefore \angle B &= 116^\circ - \angle A \\ &= (116 - 70)^\circ \\ &= 46^\circ\end{aligned}$$

Also, in ΔABC :

$$\angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 70^\circ + 46^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 64^\circ$$

Solution 8: Let $\angle A$, $\angle B$ and $\angle C$ are three angles of a triangle, then

$$\text{Let } \angle A = \angle B \text{ and } \angle C = \angle A + 18^\circ$$

Then,

$$\angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow \angle A + \angle A + \angle A + 18^\circ = 180^\circ$$

$$\Rightarrow 3\angle A = 162^\circ$$

$$\Rightarrow \angle A = 54^\circ$$

Since,

$$\angle A = \angle B$$

$$\Rightarrow \angle B = 54^\circ$$

$$\begin{aligned}\therefore \angle C &= \angle A + 18^\circ \\ &= (54 + 18)^\circ \\ &= 72^\circ\end{aligned}$$

Solution 9: Let the smallest angle of the triangle be $\angle C$ and another two angles be $\angle B$ and $\angle A$, then

$$\angle A = 2\angle C \quad \dots(1)$$

$$\angle B = 3\angle C. \quad \dots(2)$$

$$\angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 2\angle C + 3\angle C + \angle C = 180^\circ \quad [\text{From (1) and (2)}]$$

$$\Rightarrow 6\angle C = 180^\circ$$

$$\Rightarrow \angle C = 30^\circ$$

$$\therefore \angle A = 2\angle C = 2 \times 30^\circ = 60^\circ$$

$$\angle B = 3\angle C = 3 \times 30^\circ = 90^\circ$$

Solution 10:

Let ABC be a triangle right-angled at B and another two angles be $\angle A$ and $\angle C$, we can write

$$\angle B = 90^\circ \quad \dots(1)$$

$$\angle A = 53^\circ \quad \dots(2)$$

$$\therefore \angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 53^\circ + 90^\circ + \angle C = 180^\circ \quad [\text{From (1) and (2)}]$$

$$\Rightarrow \angle C = 37^\circ$$

$$\therefore \angle A = 53^\circ, \angle B = 90^\circ \text{ and } \angle C = 37^\circ.$$

Solution 11: Let ABC be a triangle and $\angle A$, $\angle B$ and $\angle C$ are three angles.

$$\text{Then, } \angle A = \angle B + \angle C \quad \dots(1)$$

$$\therefore \angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow \angle B + \angle C + \angle B + \angle C = 180^\circ \quad [\text{From equation (1)}]$$

$$\Rightarrow 2\angle B + 2\angle C = 180^\circ$$

$$\Rightarrow \angle B + \angle C = 90^\circ \quad \dots(2)$$

Now, from equation (1) and (2)

$$\Rightarrow \angle A = 90^\circ$$

This implies that the triangle is right-angled at A.

Solution 12: Let ABC be the triangle and $\angle A$, $\angle B$ and $\angle C$ are three angles of the triangle,

$$\angle A + \angle B + \angle C = 180^\circ \dots(i)$$

According to the question it can be written as

$$\angle A < \angle B + \angle C \quad \dots(ii)$$

$$\angle B < \angle A + \angle C \quad \dots(iii)$$

$$\angle C < \angle A + \angle B \quad \dots(iv)$$

Now,

Adding $\angle A$ to both sides of equation (ii)

$$2\angle A < \angle A + \angle B + \angle C$$

$$\Rightarrow 2\angle A < 180^\circ \quad [\text{From equation (i)}]$$

$$\Rightarrow \angle A < 90^\circ$$

Adding $\angle B$ to both sides of equation (iii)

$$2\angle B < \angle A + \angle B + \angle C$$

$$\Rightarrow 2\angle B < 180^\circ \quad [\text{From equation (i)}]$$

$$\Rightarrow \angle B < 90^\circ$$

Adding $\angle C$ to both sides of equation (iv)

$$2\angle C < \angle A + \angle B + \angle C$$

$$\Rightarrow 2\angle C < 180^\circ \quad [\text{From equation (i)}]$$

$$\Rightarrow \angle C < 90^\circ$$

Hence, each angle of the triangle is less than 90° .

Therefore, the triangle is acute-angled.

Solution 13: Let ABC be the triangle and $\angle A$, $\angle B$ and $\angle C$ are three angles of the triangle,

Then, according to the question, we can write

$$\angle C > \angle A + \angle B.$$

$$\Rightarrow 2\angle C > \angle A + \angle B + \angle C \quad [\text{Adding } \angle C \text{ to both sides}]$$

$$\Rightarrow 2\angle C > 180^\circ \quad [\because \angle A + \angle B + \angle C = 180^\circ]$$

$$\Rightarrow \angle C > 90^\circ$$

Since one of the angles of the triangle is greater than 90° , the triangle is obtuse-angled.

Solution 14: Side BC of triangle ABC is produced to D.

$$\therefore \angle ACD = \angle BAC + \angle ABC \quad [\text{Exterior angle property}]$$

$$\Rightarrow 128^\circ = \angle A + 43^\circ$$

$$\Rightarrow \angle A = (128 - 43)^\circ$$

$$\Rightarrow \angle A = 85^\circ$$

$$\Rightarrow \angle BAC = 85^\circ$$

Also, in triangle ABC,

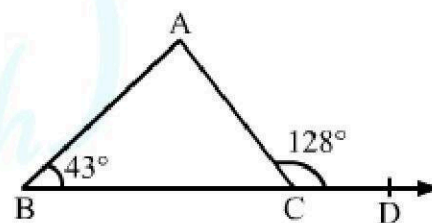
$\angle ACB$ and $\angle ACD$ form a linear pair of angles

$$\therefore \angle ACB + \angle ACD = 180^\circ$$

$$\Rightarrow \angle ACB + 128^\circ = 180^\circ$$

$$\Rightarrow \angle ACB = 180^\circ - 128^\circ$$

$$\Rightarrow \angle ACB = 52^\circ$$



Solution 15: Side BC of triangle ABC is produced to D. So, $\angle DBA$ and $\angle ABC$ form a linear pair of angles

$$\therefore \angle ABD = \angle BAC + \angle ACB$$

$$\Rightarrow \angle BAC + \angle ACB = 106^\circ \dots (i)$$

Also, side BC of triangle ABC is produced to E, $\angle ACB$ and $\angle ACE$ form a linear pair of angles

$$\angle ACE = \angle BAC + \angle ABC$$

$$\Rightarrow \angle BAC + \angle ABC = 118^\circ \dots (ii)$$

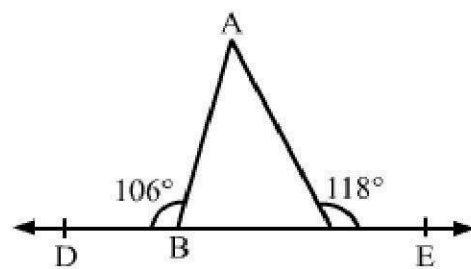
Adding (i) and (ii), we get

$$\angle BAC + \angle BAC + \angle ABC + \angle ACB = (106 + 118)^\circ$$

$$\Rightarrow (\angle BAC + \angle ABC + \angle ACB) + \angle BAC = 224^\circ$$

$$\Rightarrow 180^\circ + \angle BAC = 224^\circ \quad [\because \angle BAC + \angle ABC + \angle ACB = 180^\circ]$$

$$\Rightarrow \angle A = 44^\circ$$



$$\therefore \angle ABC = 118^\circ - \angle BAC \quad [\text{Using (ii)}]$$

$$\Rightarrow \angle ABC = (118 - 44)^\circ$$

$$\Rightarrow \angle ABC = 74^\circ$$

And,

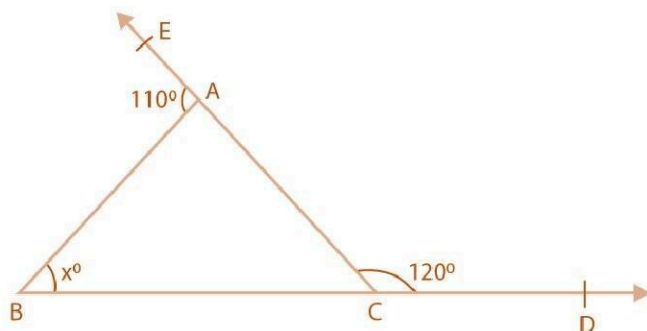
$$\angle ACB = 106^\circ - \angle BAC \quad [\text{Using (i)}]$$

$$\Rightarrow \angle ACB = (106 - 44)^\circ$$

$$\Rightarrow \angle ACB = 62^\circ$$

Solution 16:

(i)



$\angle EAB$ and $\angle BAC$ form a linear pair of angles

$$\therefore \angle EAB = \angle B + \angle C$$

$$\Rightarrow 110^\circ = x^\circ + \angle C \dots (i)$$

Also,

$$\angle ACD + \angle ACB = 180^\circ \quad [\text{linear pair}]$$

$$\Rightarrow 120^\circ + \angle ACB = 180^\circ$$

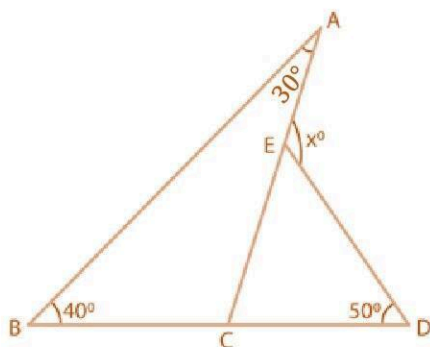
$$\Rightarrow \angle ACB = 60^\circ$$

$$\Rightarrow \angle C = 60^\circ$$

Substituting the value of $\angle C$ in (i), we get

$$x = 50$$

(ii)



From $\triangle ABC$ we have:

$$\angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 30^\circ + 40^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 110^\circ$$

$$\Rightarrow \angle ACB = 110^\circ$$

Also,

$$\angle ECB + \angle ECD = 180^\circ \quad [\text{linear pair}]$$

$$\Rightarrow 110^\circ + \angle ECD = 180^\circ$$

$$\Rightarrow \angle ECD = 70^\circ$$

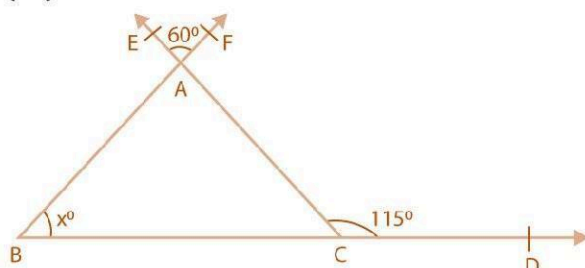
Now, in $\triangle ECD$,

$$\therefore \angle AED = \angle ECD + \angle EDC \quad [\text{exterior angle property}]$$

$$\Rightarrow x^\circ = 70^\circ + 50^\circ$$

$$\Rightarrow x^\circ = 120^\circ$$

(iii)



$$\angle ACB + \angle ACD = 180^\circ$$

[$\angle ACB$ and $\angle ACD$ are linear pair]

$$\Rightarrow \angle ACB + 115^\circ = 180^\circ$$

$$\Rightarrow \angle ACB = 65^\circ$$

Also,

$$\angle EAF = \angle BAC$$

[Vertically-opposite angles]

$$\Rightarrow \angle BAC = 60^\circ$$

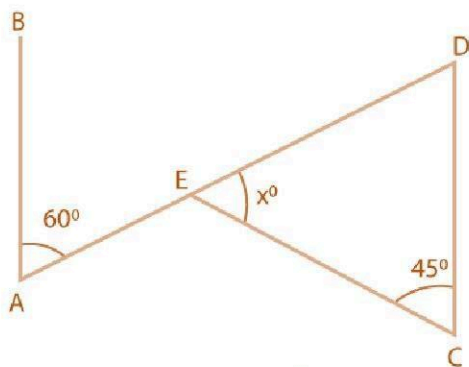
$$\therefore \angle BAC + \angle ABC = 115^\circ$$

[exterior angle property]

$$\Rightarrow 60^\circ + \angle ABC = 115^\circ$$

$$\Rightarrow x^\circ = 55^\circ$$

(iv)



$$\angle BAE = \angle CDE$$

[Alternate angles]

$$\Rightarrow \angle CDE = 60^\circ$$

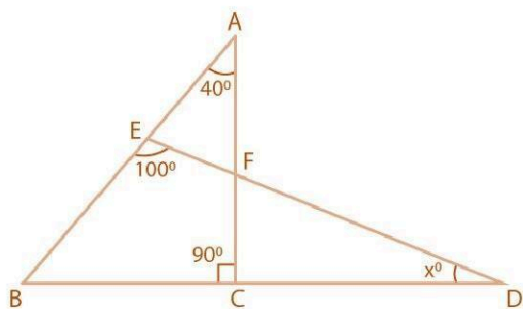
$$\therefore \angle ECD + \angle CDE + \angle CED = 180^\circ$$

[Sum of the angles of a triangle]

$$\Rightarrow 45^\circ + 60^\circ + x = 180^\circ$$

$$\Rightarrow x^\circ = 75^\circ$$

(v)



From $\triangle ABC$,

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

[Sum of the angles of a triangle]

$$\Rightarrow 40^\circ + \angle ABC + 90^\circ = 180^\circ$$

$$\Rightarrow \angle ABC = 50^\circ$$

Also, from $\triangle EBD$,

$$\angle BED + \angle EBD + \angle BDE = 180^\circ$$

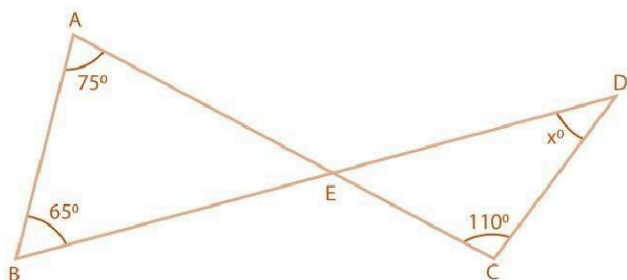
[Sum of the angles of a triangle]

$$\Rightarrow 100^\circ + 50^\circ + x^\circ = 180^\circ$$

[$\because \angle ABC = \angle EBD$]

$$\Rightarrow x^\circ = 30^\circ$$

(vi)



From $\triangle ABE$,

$$\angle BAE + \angle ABE + \angle AEB = 180^\circ$$

[Sum of the angles of a triangle]

$$\Rightarrow 75^\circ + 65^\circ + \angle AEB = 180^\circ$$

$$\Rightarrow \angle AEB = 40^\circ$$

[$\because \angle AEB = \angle CED$ Vertically-opposite angles]

$$\Rightarrow \angle CED = 40^\circ$$

Also, From $\triangle CDE$,

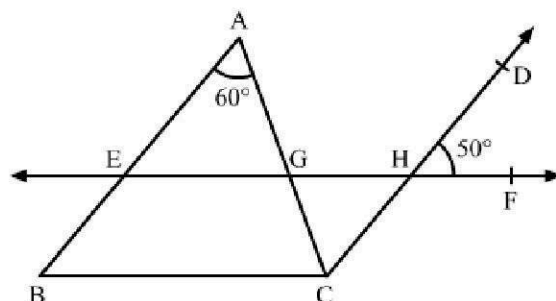
$$\angle ECD + \angle CDE + \angle CED = 180^\circ$$

[Sum of the angles of a triangle]

$$\Rightarrow 110^\circ + x + 40^\circ = 180^\circ$$

$$\Rightarrow x^\circ = 30^\circ$$

Solution 17:



Given: $AB \parallel CD$, $EF \parallel BC$, $\angle BAC = 60^\circ$ and $\angle DHF = 50^\circ$

Since, $AB \parallel CD$ and AC is a transversal

$$\therefore \angle ACD = \angle BAC = 60^\circ$$

(Pair of alternate angles)

$$\text{Also, } \angle GCH = \angle ACD = 60^\circ$$

Now, $\angle GHC = \angle DHF = 50^\circ$

(Vertically opposite angles)

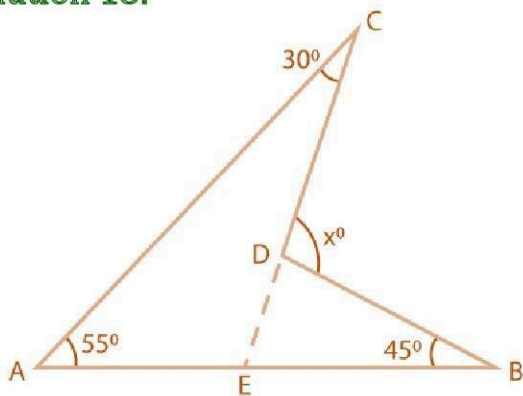
In $\triangle GCH$,

$$\angle AGH = \angle GCH + \angle GHC$$

[exterior angle property]

$$\Rightarrow \angle AGH = 60^\circ + 50^\circ = 110^\circ$$

Solution 18:



In $\triangle BDE$,

$$\angle CDB = \angle DEB + \angle DBE$$

[exterior angle property]

$$\Rightarrow x^\circ = \angle CEB + 45^\circ \quad \dots(i)$$

[$\angle DEB = \angle CEB$]

In $\triangle AEC$,

$$\angle CEB = \angle CAB + \angle ACE$$

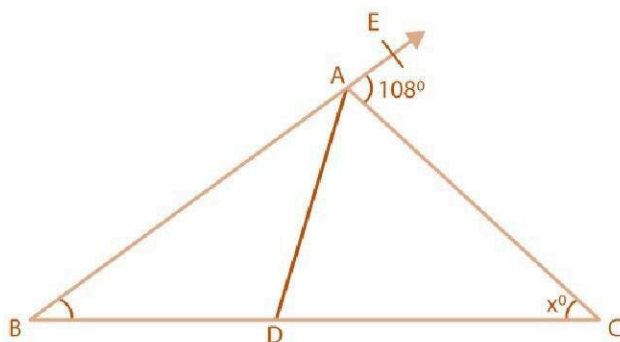
$$\Rightarrow \angle CEB = 55^\circ + 30^\circ \quad \dots(ii)$$

Adding equation (i) and (ii), we get

$$x^\circ + \angle CEB = \angle CEB + 45^\circ + 55^\circ + 30^\circ$$

$$\Rightarrow x^\circ = 130^\circ$$

Solution 19:



Let $\angle BAD$ and $\angle DAC$ are y and $3y$ respectively.

$$\therefore \angle BAD + \angle DAC = \angle BAC$$

$$\Rightarrow 3y + y = \angle BAC$$

$$\Rightarrow \angle BAC = 4y \quad \dots(i)$$

$$\text{Now, } \angle BAC + \angle CAE = 180^\circ$$

[$\because \angle BAC$ and $\angle CAE$ form linear pair]

$$\Rightarrow \angle BAC + 108^\circ = 180^\circ$$

$$\Rightarrow \angle BAC = 72^\circ$$

$$\Rightarrow 4y = 72^\circ$$

[From (i)]

$$\Rightarrow y = 18^\circ$$

$$\therefore \angle BAD = 18^\circ \quad \text{and} \quad \angle DAC = 3 \times 18^\circ = 54^\circ$$

$$\text{Now, } \angle BAD = \angle ABC = 18^\circ$$

[$\because AD = DB$]

In $\triangle ABC$,

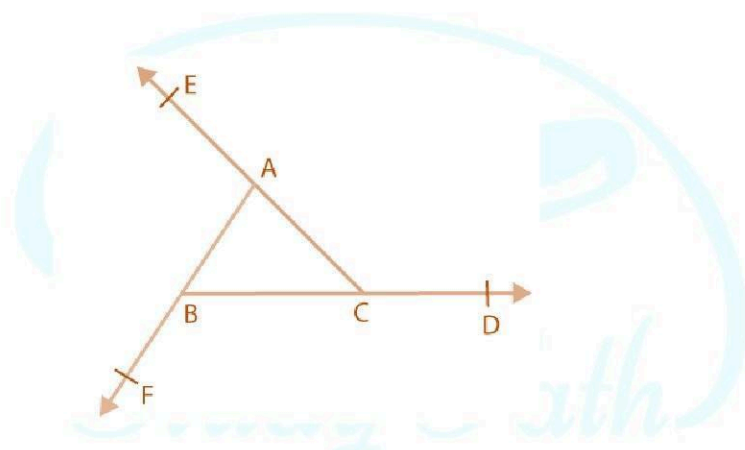
$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

[Sum of the angles of a triangle]

$$\Rightarrow 72^\circ + 18^\circ + x^\circ = 180^\circ$$

$$\Rightarrow x^\circ = 90^\circ$$

Solution 20:



$\triangle ABC$ in which AB , BC and CA are produced to points D , E and F .

Therefore,

$$\angle ACD = \angle A + \angle B \quad \dots (i)$$

$$\angle BAE = \angle B + \angle C \quad \dots (ii)$$

$$\angle CBF = \angle A + \angle C \quad \dots (iii)$$

By adding equation (i), (ii) and (iii), we get

$$\angle ACD + \angle BAE + \angle CBF = 2(\angle A + \angle B + \angle C)$$

$$\Rightarrow \angle ACD + \angle BAE + \angle CBF = 2 \times 180^\circ \quad [\text{sum of interior angles of a triangle is } 180^\circ]$$

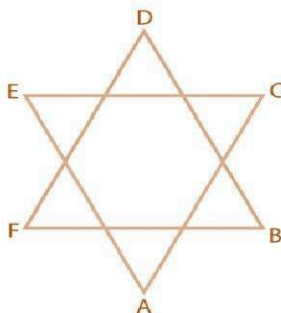
$$\Rightarrow \angle ACD + \angle BAE + \angle CBF = 360^\circ$$

$$\Rightarrow \angle ACD + \angle BAE + \angle CBF = 4 \times 90^\circ$$

$$\Rightarrow \angle ACD + \angle BAE + \angle CBF = 4 \text{ right angles}$$

Hence, the sum of the exterior angles so formed is equal to four right angles.

Solution 21:



In $\triangle ACE$,

$$\angle A + \angle C + \angle E = 180^\circ \quad \dots(i) \quad [\text{Sum of the angles of a triangle}]$$

In $\triangle BDF$,

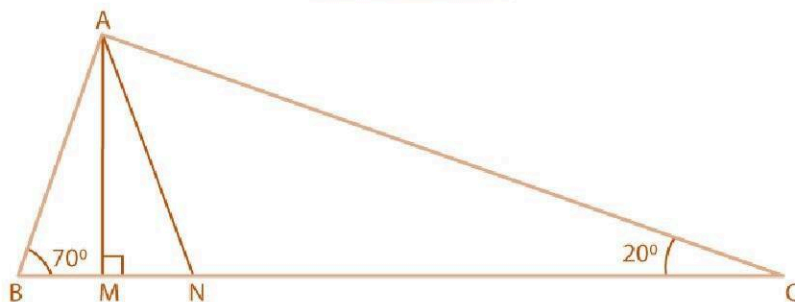
$$\angle B + \angle D + \angle F = 180^\circ \quad \dots(ii) \quad [\text{Sum of the angles of a triangle}]$$

Adding (i) and (ii), we get

$$\angle A + \angle C + \angle E + \angle B + \angle D + \angle F = (180 + 180)^\circ$$

$$\Rightarrow \angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 360^\circ$$

Solution 22:



In $\triangle ABC$,

$$\angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow \angle A + 70^\circ + 20^\circ = 180^\circ$$

$$\Rightarrow \angle A = 90^\circ$$

$$\Rightarrow \frac{1}{2} \angle A = 45^\circ$$

$$\Rightarrow \angle BAN = 45^\circ \quad [\text{AN is the bisector of } \angle A]$$

In $\triangle ABM$,

$$\angle ABM + \angle AMB + \angle BAM = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 70^\circ + 90^\circ + \angle BAM = 180^\circ$$

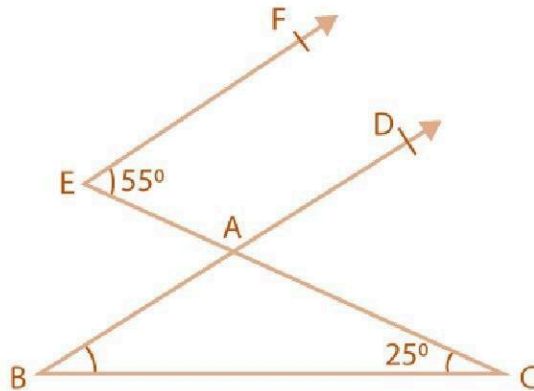
$$\Rightarrow \angle BAM = 20^\circ$$

$$\therefore \angle MAN = \angle BAN - \angle BAM$$

$$\Rightarrow \angle MAN = 45^\circ - 20^\circ$$

$$\Rightarrow \angle MAN = 25^\circ$$

Solution 23:



In the given figure, $EF \parallel BD$ and CE is the transversal.

$$\therefore \angle CAD = \angle AEF \quad (\text{Pair of corresponding angles})$$

$$\Rightarrow \angle CAD = 55^\circ$$

In $\triangle ABC$,

$$\angle CAD = \angle ABC + \angle ACB \quad (\text{Exterior angle property})$$

$$\Rightarrow 55^\circ = \angle ABC + 25^\circ$$

$$\Rightarrow \angle ABC = 55^\circ - 25^\circ$$

$$\Rightarrow \angle ABC = 30^\circ$$

Thus, the measure of $\angle ABC$ is 30° .

Solution 24:

In $\triangle ABC$,

$$\angle A + \angle B + \angle C = 180^\circ \quad \dots(i)$$

And

$$\angle A : \angle B : \angle C = 3 : 2 : 1$$

It can also be written as

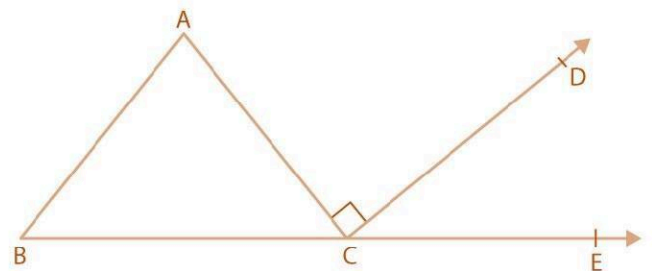
$$\angle A = 3x, \angle B = 2x \text{ and } \angle C = x$$

Now, from equation (1)

$$3x + 2x + x = 180^\circ$$

$$\Rightarrow 6x = 180^\circ$$

$$\Rightarrow x = 30^\circ$$



$$\therefore \angle A = 3x = 3 \times 30^\circ = 90^\circ$$

$$\angle B = 2x = 2 \times 30^\circ = 60^\circ$$

$$\angle C = x = 30^\circ$$

BC of triangle ABC is produced to E. So, by exterior angle is equal to the sum of two opposite interior angles

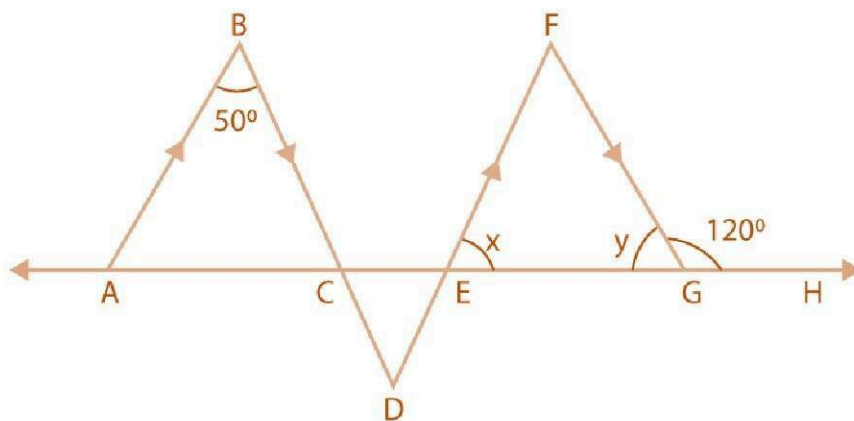
$$\therefore \angle ACE = \angle A + \angle B$$

$$\Rightarrow \angle ACD + \angle ECD = (90 + 60)^\circ$$

$$\Rightarrow 90^\circ + \angle ECD = 150^\circ$$

$$\Rightarrow \angle ECD = 60^\circ$$

Solution 25:



From the figure we have $\angle FGH$ and $\angle FGE$ form a linear pair of angles

$$\therefore \angle FGH + \angle FGE = 180^\circ$$

$$\Rightarrow 120^\circ + y = 180^\circ$$

$$\Rightarrow y = 180^\circ - 120^\circ$$

$$\Rightarrow y = 60^\circ$$

Now,

$AB \parallel DF$ and BD is the transversal. So, $\angle ABC$ and $\angle CDE$ are alternate angles

$$\therefore \angle ABC = \angle CDE = 50^\circ$$

Also,

$BC \parallel FE$ and DF is the transversal. So, $\angle EFG$ and $\angle CDE$ are alternate angles

$$\therefore \angle EFG = \angle CDE = 50^\circ$$

In $\triangle EFG$,

$$\angle FGH = \angle EFG + \angle FEG \quad [\text{Exterior angle property}]$$

$$\Rightarrow 120^\circ = 50^\circ + x$$

$$\Rightarrow x = 120^\circ - 50^\circ$$

$$\Rightarrow x = 70^\circ$$

Thus, the values of x and y are 70° and 60° , respectively.

Solution 26:

It is given that, $AB \parallel CD$ and EF is a transversal.

$$\therefore \angle EFD = \angle AEF \quad (\text{Pair of alternate angles})$$

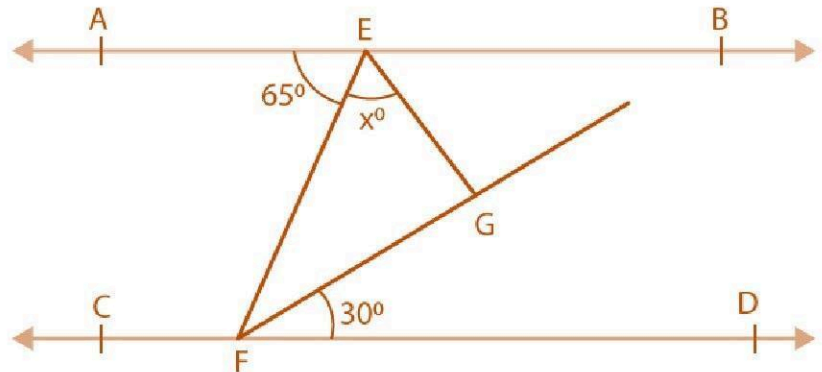
$$\Rightarrow \angle EFD = 65^\circ$$

$$\Rightarrow \angle EFG + \angle GFD = 65^\circ$$

$$\Rightarrow \angle EFG + 30^\circ = 65^\circ$$

$$\Rightarrow \angle EFG = 65^\circ - 30^\circ$$

$$\Rightarrow \angle EFG = 35^\circ$$



In $\triangle EFG$,

$$\angle EFG + \angle GEF + \angle EGF = 180^\circ \quad (\text{Angle sum property})$$

$$\Rightarrow 35^\circ + x + 90^\circ = 180^\circ$$

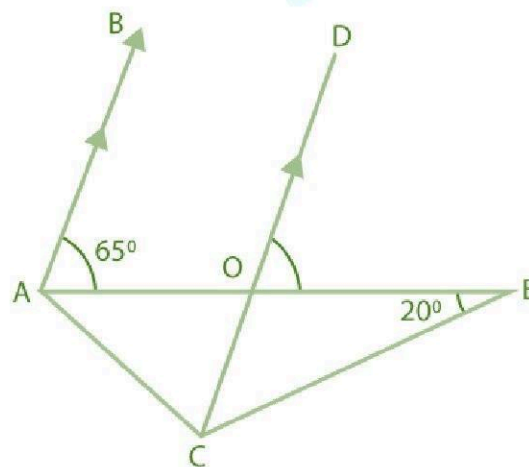
$$\Rightarrow 125^\circ + x = 180^\circ$$

$$\Rightarrow x = 180^\circ - 125^\circ$$

$$\Rightarrow x = 55^\circ$$

Thus, the value of x is 55° .

Solution 27:



In the given figure, $AB \parallel CD$ and AE is the transversal.

$$\therefore \angle DOE = \angle BAE \quad (\text{Pair of corresponding angles})$$

$$\Rightarrow \angle DOE = 65^\circ$$

In $\triangle COE$,

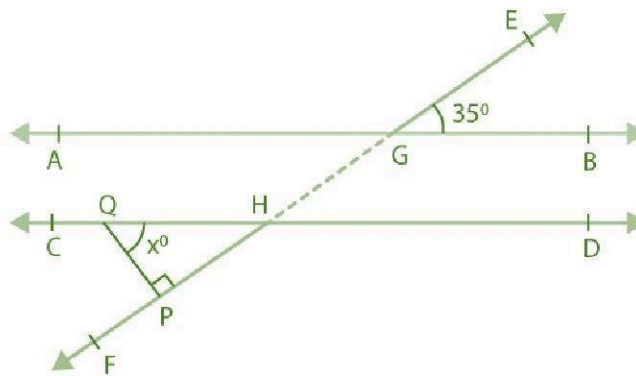
$\angle DOE = \angle OEC + \angle ECO$ (Exterior angle of a triangle is equal to the sum of two opposite interior angles)

$$\Rightarrow 65^\circ = 20^\circ + \angle ECO$$

$$\Rightarrow \angle ECO = 65^\circ - 20^\circ = 45^\circ$$

Thus, the measure of $\angle ECO$ is 45° .

Solution 28:



In the given figure, $AB \parallel CD$ and EF is a transversal.

$\therefore \angle PHQ = \angle EGB$ (Pair of alternate exterior angles)

$$\Rightarrow \angle PHQ = 35^\circ$$

In $\triangle PHQ$,

$\angle PHQ + \angle QPH + \angle PQH = 180^\circ$ (Angle sum property)

$$\Rightarrow 35^\circ + 90^\circ + x = 180^\circ$$

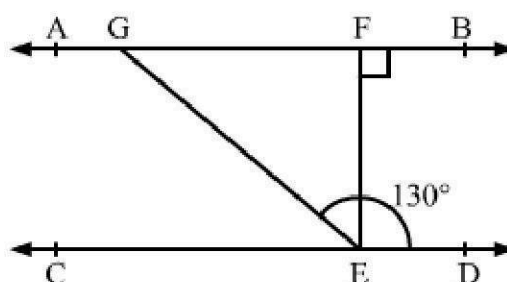
$$\Rightarrow 125^\circ + x = 180^\circ$$

$$\Rightarrow x = 180^\circ - 125^\circ$$

$$\Rightarrow x = 55^\circ$$

Thus, the measure of $\angle PQH$ is 55° .

Solution 29:



In the given figure, $AB \parallel CD$ and GE is the transversal.

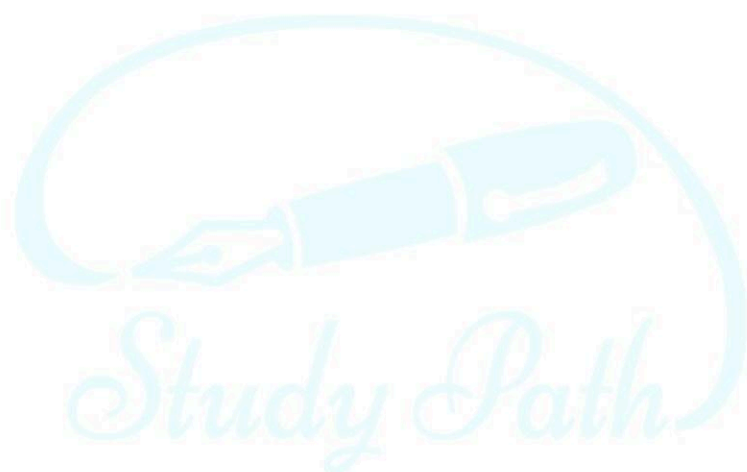
$\therefore \angle GED + \angle EGF = 180^\circ$ (Sum of adjacent interior angles on the same side of the transversal is supplementary)

$$\Rightarrow 130^\circ + \angle EGF = 180^\circ$$

$$\Rightarrow \angle EGF = 180^\circ - 130^\circ$$

$$\Rightarrow \angle EGF = 50^\circ$$

Thus, the measure of $\angle EGF$ is 50° .



Multiple Choice Questions (MCQ)

Solution 1: (b) 4 : 3 : 2

LCM of 3, 4 and 6 = 12

$$3\angle A = 4\angle B = 6\angle C \quad (\text{Given})$$

Dividing throughout by 12, we get

$$\frac{3\angle A}{12} = \frac{4\angle B}{12} = \frac{(6\angle C)}{12}$$

$$\Rightarrow \frac{\angle A}{4} = \frac{\angle B}{3} = \frac{\angle C}{2}$$

$$\text{Let } \frac{\angle A}{4} = \frac{\angle B}{3} = \frac{\angle C}{2} = k, \text{ where } k \text{ is some constant}$$

$$\text{Then, } \angle A = 4k, \angle B = 3k, \angle C = 2k$$

$$\therefore \angle A : \angle B : \angle C = 4k : 3k : 2k = 4 : 3 : 2$$

Hence, the correct answer is option (b).

Solution 2: (c) 53°

$$\angle A - \angle B = 42^\circ$$

$$\Rightarrow \angle A = \angle B + 42^\circ$$

$$\angle B - \angle C = 21^\circ$$

$$\Rightarrow \angle C = \angle B - 21^\circ$$

In $\triangle ABC$,

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle B + 42^\circ + \angle B + \angle B - 21^\circ = 180^\circ$$

$$\Rightarrow 3\angle B = 159$$

$$\Rightarrow \angle B = 53^\circ$$

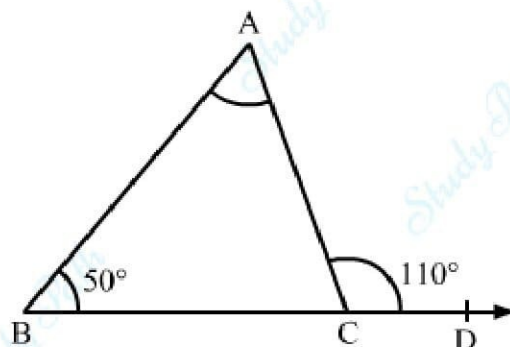
Solution 3: (b) 60°

$$\angle ACD = \angle B + \angle A \quad (\text{Exterior angle property})$$

$$\Rightarrow 110^\circ = 50^\circ + \angle A$$

$$\Rightarrow \angle A = 60^\circ$$

Hence, the correct answer is option (b).



Solution 4: (d) 75°

We have:

$$\therefore \angle ABD + \angle ABC = 180^\circ \quad [\because DE \text{ is a straight line}]$$

$$\Rightarrow 125^\circ + \angle ABC = 180^\circ$$

$$\Rightarrow \angle ABC = 55^\circ$$

Also,

$$\angle ACE + \angle ACB = 180^\circ$$

$$\Rightarrow 130^\circ + \angle ACB = 180^\circ$$

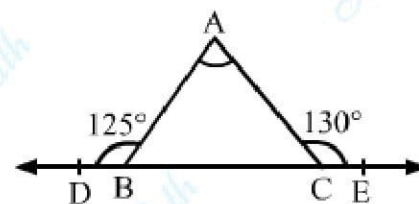
$$\Rightarrow \angle ACB = 50^\circ$$

$$\therefore \angle BAC + \angle ABC + \angle ACB = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow \angle BAC + 55^\circ + 50^\circ = 180^\circ$$

$$\Rightarrow \angle BAC = 75^\circ$$

$$\Rightarrow \angle A = 75^\circ$$



Solution 5: (a) 65°

We have:

$$\angle ABD + \angle ABC = 180^\circ \quad [\because CBD \text{ is a straight line}]$$

$$\Rightarrow 110^\circ + \angle ABC = 180^\circ$$

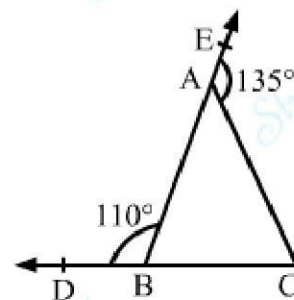
$$\Rightarrow \angle ABC = 70^\circ$$

Side AB of triangle ABC is produced to E.

$$\therefore \angle CAE = \angle ABC + \angle ACB$$

$$\Rightarrow 135^\circ = 70^\circ + \angle ACB$$

$$\Rightarrow \angle ACB = 65^\circ$$



Solution 6: (d) 360°

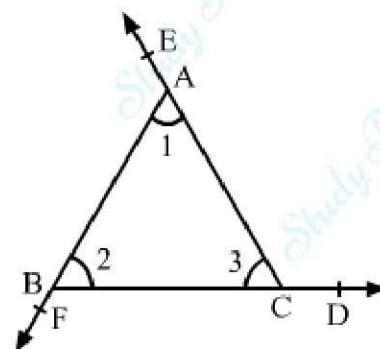
We have:

$$\angle 1 + \angle BAE = 180^\circ \quad \dots(i)$$

$$\angle 2 + \angle CBF = 180^\circ \quad \dots(ii) \text{ and}$$

$$\angle 3 + \angle ACD = 180^\circ \quad \dots(iii)$$

Adding (i), (ii) and (iii), we get



$$:(\angle 1 + \angle 2 + \angle 3) + (\angle BAE + \angle CBF + \angle ACD) = 540^\circ$$

$$\Rightarrow 180^\circ + \angle BAE + \angle CBF + \angle ACD = 540^\circ \quad [\because \angle 1 + \angle 2 + \angle 3 = 180^\circ]$$

$$\Rightarrow \angle BAE + \angle CBF + \angle ACD = 360^\circ$$

Solution 7: (b) 25

In the given figure, $\angle CAD = \angle EAF$ (Vertically opposite angles)

$$\therefore \angle CAD = 30^\circ$$

In $\triangle ABD$,

$$\angle ABD + \angle BAD + \angle ADB = 180^\circ \quad (\text{Angle sum property})$$

$$\Rightarrow (x + 10)^\circ + (x^\circ + 30^\circ) + 90^\circ = 180^\circ$$

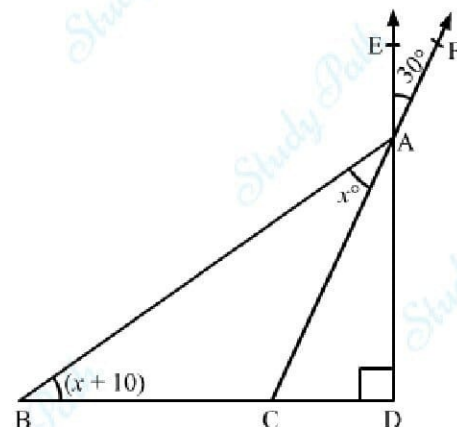
$$\Rightarrow 2x^\circ + 130^\circ = 180^\circ$$

$$\Rightarrow 2x^\circ = 180^\circ - 130^\circ = 50^\circ$$

$$\Rightarrow x = 25$$

Thus, the value of x is 25.

Hence, the correct answer is option (b).



Solution 8: (a) $x + y - 180$

In the given figure, $\angle ABF + \angle ABC = 180^\circ$ (Linear pair of angles)

$$\therefore x^\circ + \angle ABC = 180^\circ$$

$$\Rightarrow \angle ABC = 180^\circ - x^\circ \quad \dots(1)$$

Also, $\angle ACG + \angle ACB = 180^\circ$ (Linear pair of angles)

$$\therefore y^\circ + \angle ACB = 180^\circ$$

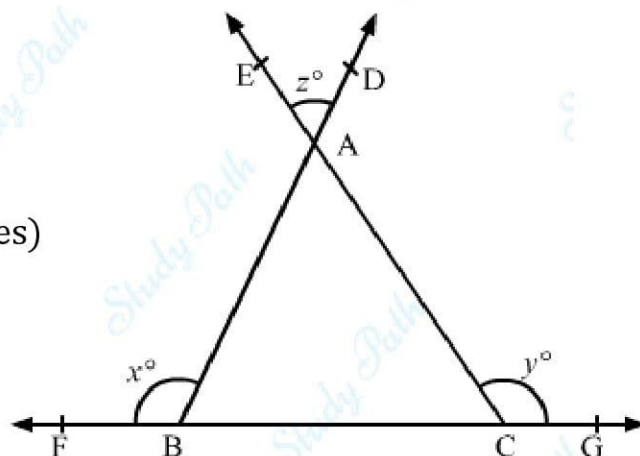
$$\Rightarrow \angle ACB = 180^\circ - y^\circ \quad \dots(2)$$

Also, $\angle BAC = \angle DAE = z^\circ$ (Vertically opposite angles) $\dots(3)$

In $\triangle ABC$,

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ \quad (\text{Angle sum property})$$

$$\therefore z^\circ + 180^\circ - x^\circ + 180^\circ - y^\circ = 180^\circ \quad [\text{Using (1), (2) and (3)}]$$



$$\Rightarrow z = x + y - 180$$

Hence, the correct answer is option (a).

Solution 9: (b) 230°

In the given figure, $\angle BOD = \angle COA$ (Vertically opposite angles)

$$\therefore \angle BOD = 40^\circ \quad \dots(i)$$

In $\triangle ACO$,

$\angle OAE = \angle OCA + \angle COA$ (Exterior angle of a triangle is equal to the sum of two opposite interior angles)

$$\Rightarrow x^\circ = 80^\circ + 40^\circ$$

$$\Rightarrow x^\circ = 120^\circ \quad \dots(ii)$$

In $\triangle BDO$,

$\angle DBF = \angle BDO + \angle BOD$ (Exterior angle of a triangle is equal to the sum of two opposite interior angles)

$$\Rightarrow y^\circ = 70^\circ + 40^\circ$$

$$\Rightarrow y^\circ = 110^\circ \quad [\text{Using (i)}] \quad \dots(iii)$$

Adding (ii) and (iii), we get

$$x^\circ + y^\circ = 120^\circ + 110^\circ = 230^\circ$$

Hence, the correct answer is option (b).

Solution 10: (a) 60°

Let $\angle A = (3x)^\circ$, $\angle B = (2x)^\circ$ and $\angle C = x^\circ$

Then,

$$3x + 2x + x = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 6x = 180^\circ$$

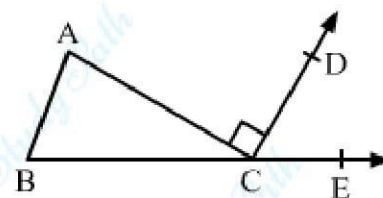
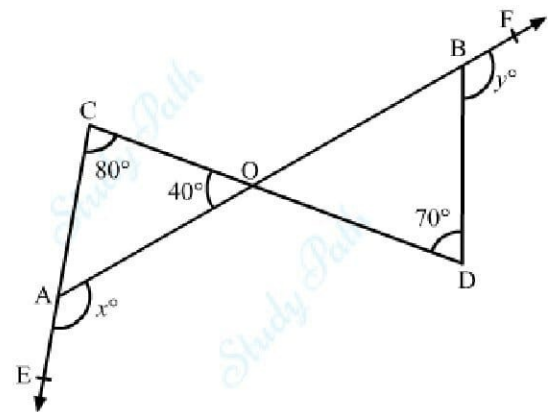
$$\Rightarrow x = 30^\circ$$

Hence, the angles are

$$\angle A = 3 \times 30^\circ = 90^\circ, \angle B = 2 \times 30^\circ = 60^\circ \text{ and } \angle C = 30^\circ$$

Side BC of triangle ABC is produced to E.

$$\therefore \angle ACE = \angle A + \angle B$$



$$\Rightarrow \angle ACD + \angle ECD = 90^\circ + 60^\circ$$

$$\Rightarrow 90^\circ + \angle ECD = 150^\circ$$

$$\Rightarrow \angle ECD = 60^\circ$$

Hence, the correct answer is option (a).

Solution 11: (c) 115°

In $\triangle ABC$, we have:

$$\angle A + \angle B + \angle C = 180^\circ \quad [\text{Sum of the angles of a triangle}]$$

$$\Rightarrow 50^\circ + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle B + \angle C = 130^\circ$$

$$\Rightarrow \frac{1}{2} \angle B + \frac{1}{2} \angle C = 65^\circ \quad \dots(i)$$

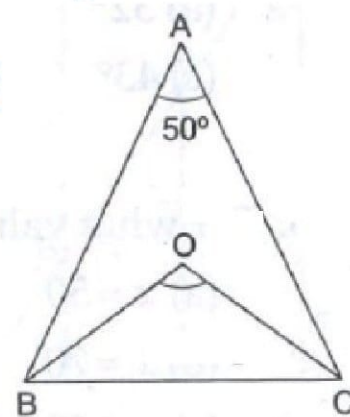
In $\triangle OBC$, we have:

$$\angle OBC + \angle OCB + \angle BOC = 180^\circ$$

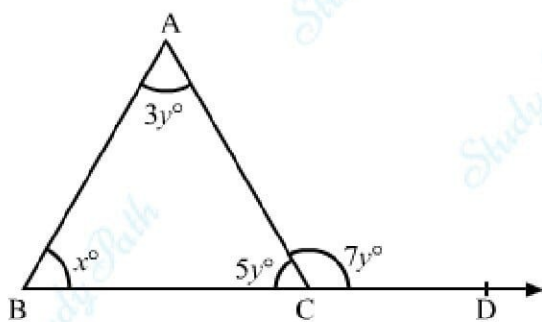
$$\Rightarrow \frac{1}{2} \angle B + \frac{1}{2} \angle C + \angle BOC = 180^\circ \quad [\text{Using (i)}]$$

$$\Rightarrow 65^\circ + \angle BOC = 180^\circ$$

$$\Rightarrow \angle BOC = 115^\circ$$



Solution 12: (a) 60



In the given figure, $\angle ACB + \angle ACD = 180^\circ$ (Linear pair of angles)

$$\therefore 5y^\circ + 7y^\circ = 180^\circ$$

$$\Rightarrow 12y^\circ = 180^\circ$$

$$\Rightarrow y = 15 \quad \dots(i)$$

In $\triangle ABC$,

$$\angle A + \angle B + \angle ACB = 180^\circ \quad (\text{Angle sum property})$$

$$\therefore 3y^\circ + x^\circ + 5y^\circ = 180^\circ$$

$$\Rightarrow x^\circ + 8y^\circ = 180^\circ$$

$$\Rightarrow x^\circ + 8 \times 15^\circ = 180^\circ \quad [\text{Using (i)}]$$

$$\Rightarrow x^\circ + 120^\circ = 180^\circ$$

$$\Rightarrow x^\circ = 180^\circ - 120^\circ$$

$$\Rightarrow x^\circ = 60^\circ$$

Thus, the value of x is 60.

Hence, the correct answer is option (a).